

Chrysanthi Kosyfaki

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ABOUT ME

My research interests are in the area of graph analytics with a particular focus on a new class of networks called Temporal Interaction Networks (TINs). In these networks, a quantity (flow) transfers from one vertex to another within a time window. TINs can model dynamic systems where entities exchange quantities, such as financial transactions, transportation flows, or communication data. Over the past few years, I have been developing efficient solutions to optimize classical problems (e.g., max-flow computation) by introducing the temporal dimension.

Currently, I am working on provenance tracking for streaming systems, where understanding how outputs are derived from inputs is critical for debugging, auditing, and explainability. I am developing how-much provenance, a new model that reports how much data each source contributed to an output, or to all outputs of a sink within a time window, without tracking individual tuples. I am also working on downstream provenance, which reverses this direction: it starts from a problematic source and identifies how many outputs that source fed, over an interval of production time and along a processing path.

EMPLOYMENT

Hong Kong University of Science and Technology (HKUST), Hong Kong SAR Postdoctoral Fellow, CSE Department	2025 – present
University of Hong Kong (HKU), Hong Kong SAR Part-time Lecturer, CS Department	2025
University of Hong Kong (HKU), Hong Kong SAR Postdoctoral Fellow, CS Department	2023 – 2025
University of Hong Kong (HKU) & MTR Corporation, Hong Kong SAR Research Assistant	2022
University of Ioannina (UoI), Greece Software Engineer - Working on different research projects (ProximIoT & SmartCityBus)	2020 - 2023
University of Hong Kong (HKU), SAR Research Visitor, CS Department	2019
University of Hong Kong (HKU), SAR Research Assistant, CS Department	2018

EDUCATION

Ionian University, Greece BSc in Computer Science ◦ <u>Thesis</u>: Sentimental Analysis in Social Networks ◦ <u>Advisor</u>: Phivos Mylonas	2013 – 2017
University of Ioannina (UoI), CSE Department, Ioannina, Greece MSc in Computer Science ◦ <u>Thesis</u>: Flow Motifs in Interaction Networks	2017 – 2018

- Advisor: Nikos Mamoulis

University of Ioannina (UoI), CSE Department, Ioannina, Greece
Ph.D in Computer Science

2019 – 2023

- Thesis: Flow Analytics in Large Graphs
- Advisor: Nikos Mamoulis

ACADEMIC SERVICE

Organization Committees

Co-chair for TKDE posters @ICDE
Publicity chair for @MDM 2026

2025-2026
2026

Conference Program Committees

VLDB
ICDE
SIGSPATIAL
PAKDD

2024-2027
2025-2027
2025
2022-23

Journal Reviewing Activity

VLDBJ
TKDE

2024-2026
2024-2026

External Reviewer

ICDE
VLDB
SIGMOD
EDBT
KDD

2019-2024
2019-2023
2021-2023
2018-2020
2019

Student Volunteer

VLDB
EDBT

2020
2023

AWARDS

Christine Collet EDBT/ICDT Student Participation Award 2019

2019

TEACHING EXPERIENCE

The University of Hong Kong (HKU), CS Department, Hong Kong SAR
Course Title: Network Data Analytics

Spring 2025

- Instructor

University of Ioannina (UoI), CSE Department, Ioannina, Greece
Course Title: Complex Data Management

2019-2023

- Teaching Assistant

University of Ioannina (UoI), CSE Department, Ioannina, Greece
Course Title: Object Oriented Programming

Spring 2018

- Teaching Assistant

University of Ioannina (UoI), CSE Department, Ioannina, Greece
Course Title: Introduction to Programming

2017-2022

- Teaching Assistant

HIGHLIGHTS

Temporal Interaction Networks (TINs) Temporal Interaction Networks (TINs) are a graph-based formalism for modeling time-varying data flows. Specifically, this type of network models dynamic systems where entities exchange quantities, such as financial transactions, transportation flows, or communication data. They can be used in a variety of problems; for computing the max-flow quantity from a source to a sink vertex to tracking the origin of a transferred quantity among the vertices (provenance).

BEACON - A Benchmark for Efficient and Accurate Counting of Subgraphs BEACON is a scalable and standardized dataset designed to evaluate ML-based techniques. It provides a unified dataset with ground truth subgraphs and different types of graphs (e.g., directed and undirected, weighted and unweighted).

SKILLS

Programming Languages

C, C++, Java, Python

Environments

MATLAB, Octave, QGIS, Neo4j

Operating Systems

Windows, MacOS, Linux

PUBLICATIONS

Stop Tracking, Start Counting: Downstream Provenance for Streaming Dataflows C.Kosyfaki and A. Simitsis - @CIDR (under review)	2027
Enhancing Constrained Shortest Path Resilience with Approximate Skyline Board Index N.Shabuerjiang, Ziyi Liu, Lei Li, C.Kosyfaki , H. Huang, A. Zhou, W. Hua, and X.Zhou - @VLDBJ (revision)	2026
An (Updated) Overview of Data Provenance: Concepts, Challenges and Opportunities C. Kosyfaki , P. Groth - @SIGMOD (tutorial)	2026
Multigranularity Spatiotemporal Flow Patterns C. Kosyfaki , N. Mamoulis, R. Cheng, B. Kao @Geoinformatica	2026
Toward Temporal Attribution Analytics in Dataflows C. Kosyfaki , R. Zhang, N. Mamoulis, X. Zhou @VLDB	2026
BEACON: A Benchmark for Efficient and Accurate Counting of Subgraphs @ICDE X. Zhu, M. Najafi, C. Kosyfaki , L. VS. Lakshmanan, R. Cheng	2026
Refine or Execute? A Cost-Based Framework for Socratic Query Refinement - @arxiv R. Zhang, C. Kosyfaki , Sau Lai Yip, X. Zhou	2026
Generalized Origin-Destination-Time Flow Patterns @SSTD C.Kosyfaki , N. Mamoulis, R. Cheng, B.Kao	2025
A Sampling-based Framework for Hypothesis Testing on Large Attributed Graphs @VLDB Y. Wang, C. Kosyfaki , S. Amer-Yahia, R. Cheng	2024
SmartCityBus - A Platform for Smart Transportation Systems @WSDM G. Bouloukakakis, C. Zeginis, N. Papadakis, K. Magoutis, G. Christodoulou, C. Kosyfaki , K. Lampropoulos, N. Mamoulis	2023
Spatiotemporal Flow Patterns @arxiv C. Kosyfaki , N. Mamoulis, R. Cheng, Ben Kao	2023
Provenance in Temporal Interaction Networks @ICDE C.Kosyfaki and N. Mamoulis	2022

Flow Provenance in Temporal Interaction Networks @SIGMOD (<u>short paper</u>) C.Kosyfaki and N. Mamoulis	2021
Flow Computation in Temporal Interaction Networks @ICDE C.Kosyfaki, N. Mamoulis, E. Pitoura, P. Tsaparas	2021
Flow Motifs in Interaction Networks @EDBT C.Kosyfaki, N. Mamoulis, E. Pitoura, P. Tsaparas	2019
Flow Motifs in Complex Networks @HDMS (<u>poster contribution</u>) C.Kosyfaki	2018
The Privacy Paradox in the Context of Online Health Data Disclosure by Users @EMCIS C.Kosyfaki, N. Angelova, A. Tsohou, E. Mangos	2017

STUDENT SUPERVISION

PhD Students HKU

- Carrie Wang - working on hypothesis testing on graphs
- Xiangju Zhu - working on subgraph counting problems
- Matin Najafi - worked on subgraph counting problems, now a researcher at Huawei, Hong Kong

Bachelor Students UoI

- Ioanna Papayianni (2020) - Thesis: Developing efficient algorithms for analyzing flow patterns in large networks
- Dimitris Zervas (2021) - Thesis: Detecting the origin of transactions in the Bitcoin Network
- Sotiria Kastana (2021) - Thesis: Design and development of a synthetic indoor movement generator
- Vasileios Georgoulas (2023) - Thesis: A web application for passenger movement with public transport